

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex. By the definition of convexity,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \quad \text{for all } x, y \in \mathbb{R}, t \in [0, 1]. \quad (1)$$

We prove Jensen's inequality

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i) \quad (J)$$

for arbitrary real numbers $x_1, \dots, x_n$ and non-negative weights λ_i with $\sum_{i=1}^n \lambda_i = 1$, by mathematical induction on n .

Base case $n = 2$. Take $t = \lambda_1$, $x = x_1$ and $y = x_2$ in (1). Since $\lambda_2 = 1 - \lambda_1$,

$$f(\lambda_1 x_1 + (1 - \lambda_1)x_2) \leq \lambda_1 f(x_1) + (1 - \lambda_1)f(x_2),$$

which is precisely (J) for $n = 2$.

Inductive step. Assume that (J) holds for some integer $k \geq 2$; i.e., for any points $y_1, \dots, y_k$ and weights $\mu_1, \dots, \mu_k \geq 0$ with $\sum_{i=1}^k \mu_i = 1$,

$$f\left(\sum_{i=1}^k \mu_i y_i\right) \leq \sum_{i=1}^k \mu_i f(y_i). \quad (IH)$$

Consider $k+1$ points $x_1, \dots, x_{k+1}$ with weights $\lambda_1, \dots, \lambda_{k+1} \geq 0$ and $\sum_{i=1}^{k+1} \lambda_i = 1$. Set

$$\alpha := 1 - \lambda_{k+1} = \sum_{i=1}^k \lambda_i \in [0, 1].$$

If $\alpha = 0$, then $\lambda_{k+1} = 1$ and all other λ_i are zero. In this case

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) = f(x_{k+1}) = \lambda_{k+1} f(x_{k+1}) + \sum_{i=1}^k \lambda_i f(x_i),$$

so (J) holds trivially. Hence we may assume $\alpha > 0$.

Define

$$y := \frac{1}{\alpha} \sum_{i=1}^k \lambda_i x_i.$$

The coefficients $\frac{\lambda_i}{\alpha}$ are non-negative and sum to 1, so y is a convex combination of the first k points.

Apply (J) with $t = \alpha$, $x = y$ and $y = x_{k+1}$:

$$f(\alpha y + (1 - \alpha)x_{k+1}) \leq \alpha f(y) + (1 - \alpha)f(x_{k+1}). \quad (2)$$

Since $\alpha y = \sum_{i=1}^k \lambda_i x_i$ and $1 - \alpha = \lambda_{k+1}$,

$$f\left(\sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1}\right) = f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right). \quad (3)$$

By the inductive hypothesis applied to the convex combination that defines y ,

$$f(y) \leq \sum_{i=1}^k \frac{\lambda_i}{\alpha} f(x_i),$$

and multiplying by α yields

$$\alpha f(y) \leq \sum_{i=1}^k \lambda_i f(x_i). \quad (4)$$

Insert (??) into the right-hand side of (??) and use (??):

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i).$$

Thus Jensen's inequality holds for $k+1$.

Conclusion. The statement is true for $n = 2$ (base case) and the inductive step shows that truth for k implies truth for $k+1$. By mathematical induction, (??) holds for every integer $n \geq 2$ (the case $n = 1$ is immediate because both sides are equal).

Equality condition. From (??) we know that equality holds iff f is affine on the segment joining the two points used in (??). Consequently, equality in Jensen's inequality occurs exactly when all the points x_i lie in an interval on which f is affine; equivalently, either all the x_i are equal, or f is linear on the convex hull of $\{x_1, \dots, x_n\}$ (weights equal to zero are irrelevant). □